

AI Engineer

Step 2: Master Machine Learning

CareerCompass — Detailed Learning Notes

What this step is about

Learn the complete ML workflow and the core algorithms used in practical systems.

Core Concepts

- Follow the workflow: define problem, collect data, clean, split, train, validate, evaluate, tune and monitor.
- Understand supervised vs unsupervised learning and classification vs regression.
- Learn linear/logistic regression, trees, random forests, boosting and clustering.
- Understand leakage, overfitting, underfitting, bias/variance and evaluation metrics.

Skills You Should Be Able To Demonstrate

- Build correct train/validation/test splits.
- Choose metrics based on the business problem.
- Compare multiple models.
- Use preprocessing pipelines.
- Explain overfitting and regularization.

Hands-on Project / Practice

- Build a classification project with reproducible preprocessing and at least three candidate models.
- Include baseline, final metrics and error analysis.

Recommended References

- Scikit-learn Documentation — https://scikit-learn.org/stable/user_guide.html
- Machine Learning with Python and Scikit-Learn — <https://www.youtube.com/watch?v=hDKCxebp88A>